Topic 17C: Carboxylic acids
9. be able to identify the carboxylic acid functional group
10. understand that hydrogen bonding affects the physical properties of carboxylic acids, in relation to their boiling temperatures and solubility
11. understand that carboxylic acids can be prepared by the oxidation of alcohols or aldehydes, and the hydrolysis of nitriles
12. understand the reactions of carboxylic acids with: i lithium tetrahydridoaluminate (lithium aluminium hydride) in dry ether ii bases to produce salts iii phosphorus(V) chloride (phosphorus pentachloride) iv alcohols in the presence of an acid catalyst
13. be able to identify the acyl chloride and ester functional groups
14. understand the reactions of acyl chlorides with: i water ii alcohols iii concentrated ammonia iv amines
15. understand the hydrolysis reactions of esters, in acidic and alkaline solution
16. understand how polyesters are formed by condensation polymerisation reactions

Topic 18: Organic Chemistry III

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include investigating the reactions of different functional groups, preparing an aromatic ester such as methyl benzoate, making nylon, purifying an organic solid.

Mathematical skills that could be developed in this topic include calculating the resonance stability of benzene from thermodynamic data, calculating percentage yields.

Within this topic, students can consider how the model for benzene structure has developed in response to new evidence. By this stage, their continuing practical experience should enable them to use techniques to carry out reactions and purify products efficiently and safely.